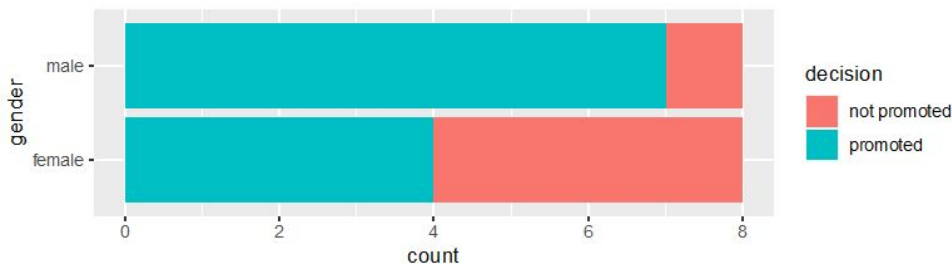


Bank Supervisors Again: Conclusions

MATH 213

The Questions

Think again about the experiment on bank supervisors who were given identical candidate files, half of which indicated the candidate was male and half of which indicated the candidate was female. Imagine that instead of working with a sample of 48 managers, the researchers experimented on only 16 managers. I made up some data for such a small experiment—with similar proportions to what we saw for the real experiment—and summarized it in the following bar graph.

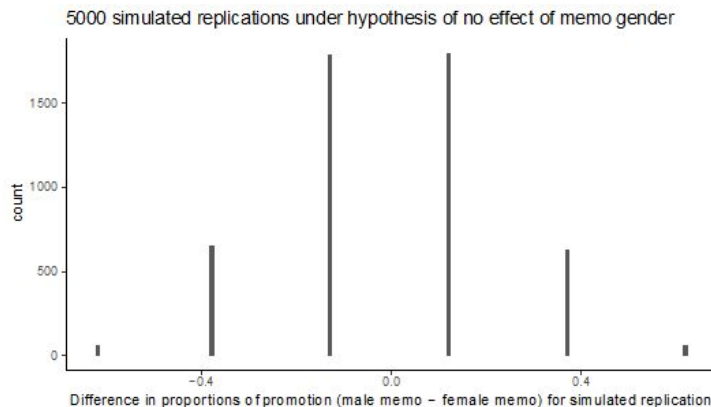


Use the bar graph to calculate the difference in proportions of those who would promote between those in the male-file group and those in the female-file group. (M-F)

$$\text{Ans: } \frac{7}{8} - \frac{4}{8} = \frac{3}{8} = 0.375$$

To continue the analysis, I evaluated the hypothesis that the sex associated with the file had no effect on the decision of any bank administration for whether to promote or not. I used R to simulate repeating the experiment (under this hypothesis) 5000 times. The following histogram summarizes these 5000 simulated replications.

What do you conclude from my 5000 simulated replications about the hypothesis that bank administrators would all have made the same recommendation regardless of the sex associated with the file? Explain your conclusion, including some numbers drawn from the histogram to support what you write. [See next slide...](#)



Bank Supervisors Again - Conclusions

Assuming that the sex associated with the file had no effect on the supervisors' decisions, the difference of proportions between those in the male-file group and those in the female-file group of 0.375 from the experiment is not so unusual. A result as or more extreme than this occurred about 700 times out of 5000 in our simulation. This means that this occurred about $700/5000 = 0.14$, about 14% of the time, which is not so unusual. In other words, the experimental difference of proportions could have occurred due to chance alone (the random assignment process), OR it could have occurred due to discrimination. We don't have enough information to be able to tell. Having only a sample size of 8 in each group means that only a few differences of proportions are even possible, and the difference of $\frac{3}{8}$ occurs fairly frequently. It would make sense to run this experiment using a larger sample size.

Note: the probability of $700/5000 = 0.14$ that we calculated is called a **p-value**.